

Biochemical Engineering ChE422
Spring 2020
Department of Chemical Engineering

422-41847 for Undergraduate

422-41848 for Graduate

Lecture Hours: Tuesday and Thursday 2:30pm – 3:45pm

Office Hours: Tuesday and Thursday 1 – 2pm

Instructor: Dr. Gang Cheng (EIB 124)

Email: gancheng@uic.edu

Prerequisites: Biochemistry, Microbiology or organic chemistry, May be taken concurrently, or by permission.

The course is aimed at senior undergraduate and graduate students and is designed to move engineers and other non-biologists past the initial language and knowledge barriers of the Biochemical Engineering field. This course consists of two parts. The first half part of this course will introduce the fundamental recombinant DNA techniques required to genetically engineer biomacromolecules, as well as the most important general molecular biology techniques, including Polymerase Chain Reaction (PCR) basics, site-directed and random mutagenesis, display techniques, DNA sequencing, DNA and protein gel electrophoresis, cutting and splicing DNA, protein expression, DNA and RNA hybridization techniques, and Western blots. Additionally, students will learn different protein design approaches, including knowledge-based design, computational protein design, and directed evolution, that are commonly used for protein engineering. Examples of engineered proteins with novel structural and functional properties are extensively discussed to illustrate how design principles are applied to real-life problems. The second half part of this course will introduce fermentation (basic principles of fermentation processes, scaling-up), product recovery and separation (cell disruption, sedimentation, filtration/centrifugation, solvent extraction, adsorption, precipitation), product purification (adsorption, ultrafiltration, crystallization) and final separation and advances in bioseparation strategies. At the end of this course, each student will be asked to design a process from the creation of the engineered strain and cell culture/fermentation to the purification of the end product for energy, biotechnology or biomedical applications and write a report.

Recommended references:

(1) Bioprocess Engineering: Basic Concepts (3rd Edition)

(2) Molecular Biology of the Cell by Bruce Alberts

(3) Biochemistry (any)

Learning outcomes

To be able to describe and understand recombinant DNA, fermentation and separation methods such as PCR, filtration and centrifugation, cell rupture, and purification techniques such as adsorption, solvent extraction, membrane filtration, chromatography (affinity, ion, hydrophobic, reverse phase), precipitation, and crystallisation. To apply fundamental principles for the separation of biopharmaceuticals via chemical engineering unit operations.

Homework:

Homework is a significant part of this course. It should represent your own effort though you may discuss problems with others. Homework that is copied between students will earn a grade of zero for both students. Homework assignments will be due 7 days after being given out at the beginning of class.

Presentation:

There will be one presentation. You need to give a 15 minutes paper critique. The paper will be assigned by the instructor.

Examinations:

There will be one midterm exam and one final exam.

Final report:

At the end of the course, you need to write a 5 pages literature review on one of the given topics.

Grading Policy:

Homework: 10%

Midterm Exam: 25%

Presentation and final report: 25%

Final Exam: 40%